ISB140/DS 110−INTRODUCTION TO STATISTICS

LECTURE NOTES

UNIT I: INTRODUCTION TO STATISTICS

1. INTRODUCTION TO STATISTICS

- ✓ Statistics is mostly used in our daily lives, be it at our workplaces our homes or in the business world.
- ✓ Hence, it's not rare to use statements like, what's the average monthly sales figure or what's our daily power consumption in the home?
- ✓ The use of statistics in our daily lives helps to improve our understanding of the world around us and sharpens our ability to make better decisions in the face of many data and uncertainty.
- ✓ For instance, through statistics **Engineers** are able to decide best quality of manufactured products.
- ✓ **Economists** can predict future economic outlook and give useful summary statistics to aid politicians make appropriate policies.

1.1. NATURE OF STATISTICS

- ✓ What is statistics? Consider the following examples:
- ✓ A pollster interested in the outcome of an upcoming election interviews a certain number of voters and makes a prediction as to who will probably win the election.
- ✓ To get an idea of the economic status of a town, an economist obtains the salaries of all workers in the town and then computes the average.
- ✓ To estimate the average age of students at KMU, a ICT student selects 1000 students, computes their average age and then uses this value as an estimate.
- ✓ All these examples make use of data or observations. An observation might be numerical, such as a person's age or non-numerical such as a person's gender (male or female).

1.2. IMPORTANCE OF STATISTICS

- ✓ Statistics is very useful and its importance in society can never be overemphasized.
- ✓ One may tend to ask, why is statistics important? What is its relevance in society?
- ✓ The following are some of the importance of statistics:
  - (a) Statistics enables you to understand about the risks associated with making a concrete decision in every fields, eg business.
  - (b) It allows you reduce the variation in the decision-making process
  - (c) Statistics provides you with methods for making better sense of the numbers used every day to describe or analyze the world we live in.

(d) Statistics helps in interpreting the meaning from statistical information, this is because Numerical information is everywhere. It can be found in Newspapers and other sources of information where we see exchange rate data, price data and all kinds of statistical information.

(e) Statistics provide you with the knowledge of statistical methods that will help you understand how decisions are made and give you a better understanding of how they affect you.

(f) Statistics provides you with the groundwork for more advanced studies in data science, machine learning, and quantitative research.

1.3. USES OF STATISTICS

- ✓ Primarily there are two basic uses of statistics, these are:

(i) To compare or describe data , that is, to visualize and summarize data (Descriptive statistics).

(ii) To make inferences about certain uncertain events, that is, to reach conclusions about a large group based on data collected from a small group as well as to make a reliable forecasts that are based on statistical models for predictions (inferential statistics).

2. STATISTICS AND TYPES OF STATISTICS

2.1. DEFINITION (STATISTICS)

- ✓ The word statistics can refer to a collection of numbers or it can refer to the science of studying collections of numbers.
- ✓ Thus, Statistics is the science of collecting, simplifying and describing data as well as making inferences (drawing conclusions) based on the analysis of data.
- ✓ In other words, statistics is the science of collecting, organizing, presenting,analyzing, and interpreting data to assist in making more effective decisions.
- ✓ Hence, it is suffice to note that there are three main tasks in statistics and these are:

(a) Collection and organization,

(b) Analysis, and

(c) Interpretation of data

2.2. TYPES OF STATISTICS

- ✓ There are two branches (or types) of statistics namely **Descriptive** and **Inferential** statistics.

A. DESCRIPTIVE STATISTICS

- ❖ Descriptive statistics consists of methods used for organising, simplifying and summarising data to give a clear effective way.

- Thus, the objective of descriptive statistics is to organize, summarize or describe the data to make it more comprehensible (clear) e.g. A list of the salaries of workers in Lusaka would be incomprehensible because of its length but the average of these salaries would give an understanding of the economic status of Lusaka residents.
- Some of the methods used in descriptive statistics include: graphs, charts, tables and calculations of various descriptive measure such as averages and percentages.
- In descriptive statistics there are various statistical measures that can be used to give descriptive analysis of data. These are measures of central tendency, measures of dispersion or spread and distribution.
- So the key descriptive statistics are those that measure central tendency of data, its dispersion and distribution.
- There are three most commonly used measures of central tendency (also known as location measures): The Mean, Median and Mode.
- These measures describe the centre, central or middle value of a group of data or a value which more accurately represents the center of the data.
- In conclusion, descriptive statistics consists of methods for organizing, displaying, and describing data by using tables, graphs, and summary measures.

B. Inferential statistics

- Inferential statistics consists of methods used to draw general conclusions (inferences) about the population based on information obtained from a sample of the population.
- In other words, Inferential Statistics consists of methods that use the data collected from a small group (sample) to draw conclusions about a larger group (population).
- For example, the ICT student who estimates the average age of KMU students based on a knowledge of the average age of some of these students uses inferential statistics.
- There are two main branches of statistical inferences, these are Estimation and hypothesis testing.
- These procedures are closely related but serve very different purposes. We shall develop skills in both the commonly used methods of estimation of population parameters and how to test hypotheses about these estimates later in the course.
- In conclusion inferential statistics consists of methods or proceedures that use sample results or information contained in a sample to help make decisions or predictions or inferences about a population parameter.
- There are two main branches of statistical inferences, namely: Estimation and hypothesis testing.

Note

1. Inferential statistics can be defined in terms of population and sample.
2. Inferential statistics is concerned with making judgements (or inferences) about a population based on the properties of some sample obtained from the population.

2.3. BASIC STATISTICAL TERMS

- ✓ It is very important to understand the meaning of some basic statistical terms that are often be used in statistics.

2.3.1. Element or member

- ✓ **Defn**: An element or member of a sample or population is a specific subject or object (for example, a person, firm, item, state, or country) about which the information is collected.

2.3.2. Population vs Sample

- ✓ **Defn**: Population is the collection of all individual items under consideration in a statistical study.
- ✓ In other words, population is defined as the entire collection of all the elements of interest.
- ✓ These elements might be people, cars, data values, etc.
- ✓ **Defn**: A sample is a collection of some of the elements obtained from the population
- ✓ In other words, a sample is a part of the population from which information is collected or it is the selection of a portion of the elements from the population.
- ✓ Using the elements in the sample, we draw conclusions about the population.

2.3.3. Parameter vs Statistics

- ✓ **Defn**: A parameter is a numerical property of a population or is a measure that describes a characteristic of a population.
- ✓ **Defn**: A statistic is a numerical property of a sample or is a measure that describes a characteristic of a sample. Numerical property means a property that can be expressed as a number.
- ✓ For example, if the problem question is: estimate the age of KMU students, then
  - (a) Population: all KMU students (specifically ages of all KMU students).
  - (b) Parameter: the average age of all KMU students.
  - (c) Statistic: the average of the ages for a sample of KMU students.

3. TYPES OF VARIABLES

- ✓ **Defn**: A variable is a characteristic of an item or individual of interest that assumes different values for different elements.
- ✓ A variable may be classified into two types, namely **quantitative variables** and **qualitative or categorical variables**. These two types of variables are explained fully below.

A. QUANTITATIVE OR NUMERICAL VARIABLES

- ✓ A quantitative variable also known as a numerical variable is a variable that can be measured numerically.
- ✓ Quantitative variables may be classified as either **Discrete variables** or **Continuous variables**.

(i) DISCRETE VARIABLES

- ❖ A discrete variable is a variable whose values are countable or it is a variable whose value is obtained by the process of counting.
- ❖ This variable will contain integer values such as $0,1,2,3,\ldots$.
- ❖ Examples of discrete varibles are: Number of voters in an election, Number of students in DS104 in the year 2025 etc..
- ❖ In other words, a discrete variable can assume only certain values with no intermediate values.

(ii) CONTINUOUS VARIABLES

- ❖ A continous variable is a variable that can assume any numerical value over a certain interval or intervals or it is a variable whose value is obtained by the process of measuring.
- ❖ Examples of countinouus variables are: Amount of rainfall, A persons weight, the waiting time for teller service at the bank etc.

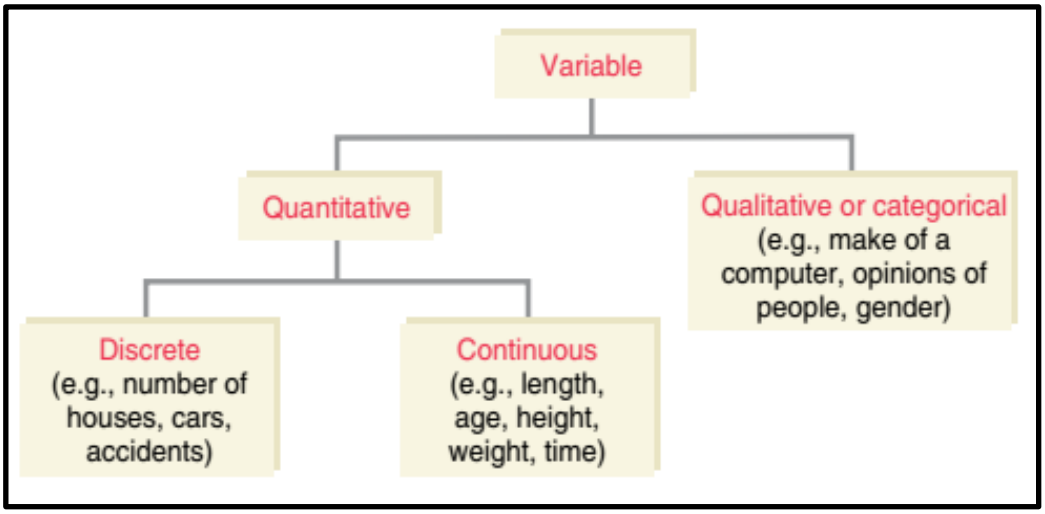
B. QUALITATIVE OR CATEGORICAL VARIABLES

- ✓ A qualitative variable also known as a categorical Variable is a variable that cannot assume a numerical value but can be classified or grouped into two or more non-numeric categories.
- ✓ For example, the status of an undergraduate university student is a qualitative variable because a student can fall into any one of four categories: **fresher, mafoursa, mathird**, or **mafifi**.
- ✓ Other examples of qualitative variables are the gender of a person, the make of a computer, the opinions of people, and the make of a car etc.

NOTE

- ✓ The classifications of the qualitative or categorical Variables are into two scales or levels of measurements namely, the **Nominal** and **Ordinal** scales of measurements. Further, explanation on both measurement scales will be given below.

THE SUMMARY OF THE DIFFERENT TYPES OF VARIABLES



4. SCALES OR LEVELS OF MEASUREMENT

- ✓ The values for variables can themselves be classified by the level of measurement, or measurement scale.
- ✓ There are essentially four (4) types of scales of measurement namely Nominal, Ordinal, Interval and Ratio.

4.1. NOMINAL SCALE

- ✓ It involves classification of items into discrete groups/categories.
- ✓ Normally, these groups are mutually exclusive, and they are also exhaustive so that all the elements are in one group.e.g., sex – male and female, marital status – married, single, divorced and widowed.
- ✓ Nominal scales are categorical since they do not belong to other groups.
- ✓ Under this scale, numbers are arbitrary nothing, they are just for purposes of identification.
- ✓ For example, under gender we may have 1=female and 2=male and under married we may have 1=single, 2=maried, 3=divorced and 4=widowed.
- ✓ Nominal scales have no magnitudes, no equal intervals and no absolute zero (no attributes at all).
- ✓ In general, a nominal scale classifies variables into distinct categories in which no ranking is implied .

NOTE

(i) **Exclusive** means that no item or element can belong to more than one group at the same time. Implying that if something is in one group, it cannot be in another. For example, Example: If you classify people based on blood type (A, B, AB, or O), a person can only belong to **one** blood type category.

(ii) **Exhaustive** means that the groups cover all possible cases or elements. Implying that nothing is left out, every item must belong to at least one group. Example: In the blood type classification, every human has a blood type (A, B, AB, or O), so the classification is **exhaustive** because there is no one left out.

4.2. ORDINAL SCALE

- ✓ Ordinal scale involves the ranking/ordering of categories or groups of particular variables in either ascending or descending order.
- ✓ Ordinal scale must be exclusive and exhaustive.
- ✓ Ordinal scale has magnitude but do not have equal intervals and absolute zero.
- ✓ Ordinal scale is more complex and informative than nominal scale.
- ✓ Ordinal scale allows comparison and establish rank-order between things or different values of a variable.
- ✓ Thus it is possible in ordinal scale to state that one value is greater or less than another value.
- ✓ In ordinal scale, the feelings of a respondent are not only classified into happy or unhappy but into very happy, happy or indifferently unhappy, very unhappy, thus enabling comparison between degree or degrees of happiness of different persons.
- ✓ Of course this type of measurement still does not allow one assess that person A is twice as happy as person B because even though a certain magnitude can be given to a feeling of person A and B, there exist no unit to measure them.
- ✓ Therefore, only one property of scale is present and that is magnitude, and there is no existence of equal interval or absolute zero.
- ✓ In general, an ordinal scale classifies values into distinct categories in which ranking is implied. e.g. student grades $(A+, A, B+, B, C+, C, D)$.

4.3. INTERVAL SCALE

- ✓ Interval scale consists of continuous measurements with no minimal size unit and are divided into fractional units/decimal places.

- ✓ An Interval scale has two properties that include Magnitude and equal intervals but no absolute zero because it is possible to get a negative value.
- ✓ An Interval scale is an improvement over both nominal and ordinal scale in the sense that a comparison between different values of a variable is made more accurate by the introduction of equal intervals or units of measurement.
- ✓ For example, after operationalizing the concept of 'employment' a person is classified as employed or unemployed on a **nominal scale**; employed full time, employed part-time on an **ordinal scale**; employed for a certain number of hours per week on a scale possessing equal intervals.
- ✓ In general an interval scale is an ordered scale in which the difference between measurements is a meaningful quantity but does not involve a true zero point.
- ✓ For example, a noontime temperature reading of 67 degrees Fahrenheit is 2 degrees warmer than a noontime reading of 65 degrees.

4.4. RATIO SCALE

- ✓ A ratio scale is an ordered scale in which the difference between the measurements involves a true zero point, as in height, weight, age, or salary measurements.
- ✓ Ratio scale possesses the three properties of magnitude, equal intervals and absolute zero.
- ✓ In a ratio scale it is possible to do division/analysis, and it can accommodate all sorts of age, weight, speed, etc.
- ✓ The table below summarizes the properties ofn the levels of measurements.

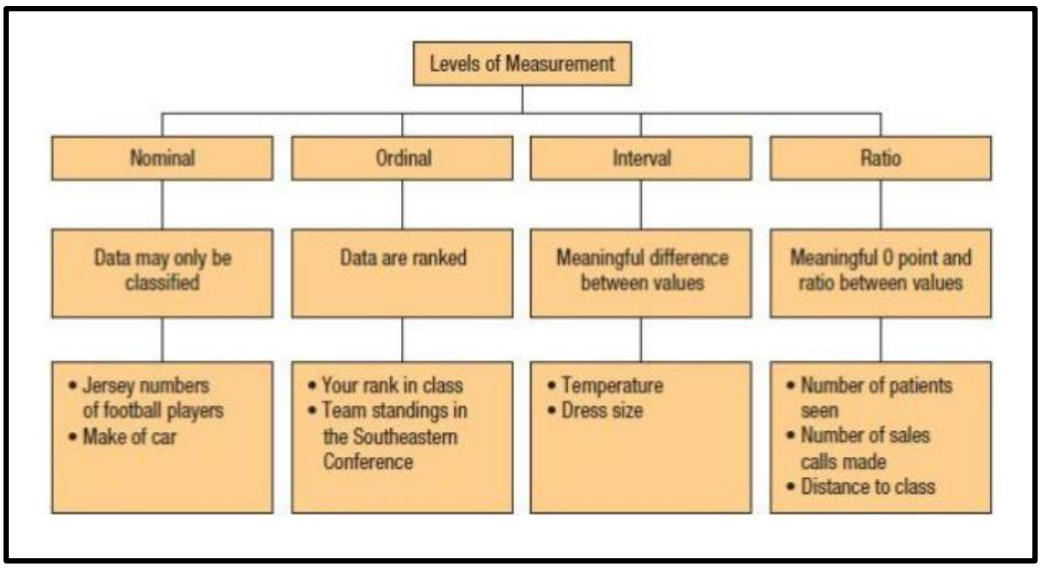
Scale	Magnitude	Equal	Absolute zero
Nominal	No	No	No
Ordinal	Yes	No	No
Interval	Yes	Yes	No
Ratio	Yes	Yes	Yes

NOTE

(i) Statisticians use the terms **nominal scale** and **ordinal scale** to describe the values for a categorical variable.

(ii) They also use the terms **interval scale** and **ratio scale** to describe numerical values.

SUMMARY OF LEVELS OF MEASUREMENT



EXAMPLE

For each of the following variables, determine whether the variable is Categorical or Numerical. If the variable is numerical, determine whether the variable is discrete or continuous. In addition, determine the measurement scale for each variable.

(a) Number of cellphones in the household.

(b) Length (in minutes) of the longest telephone call made in a month.

(c) Whether someone in home owns a WiFi capable cell phone

(d) Whether there is a high speed Internet connection in household.

ANSWERS

(a) numerical, discrete, ratio scale

(b) numerical, continuous, ratio scale

(c) categorical, nominal scale

(d) categorical, nominal scale.

5. TYPES OF DATA

- ✓ Data can be classified in several ways, however, we will give two different classifications or types of data, that is, one based on whether the data are measured on a numerical scale or not, and the other on whether the data are collected in the same time period or collected at different time periods.
- ✓ These two classifications of data include: quantitative data and qualitative data.
- ✓ **Defn**: Data is any information collected, organized and analyzed. In otherwords**,** data are the different values associated with a variable.

- ✓ **Defn**: Observation or Measurement is the value of a variable for an element
- ✓ **Defn:** A data set is a collection of observations on one or more variables.

5.1. QUANTITATIVE OR NUMERIC DATA

- ✓ Quantitative data are observations that are measured on a numerical scale or data collected on a quantitative variable.
- ✓ Just like quantitative variables, quantitative data may be classified as either **Discrete data** or **Continuous data**.
- ✓ **Defn**: Discrete data is a numerical data or variable with a countable number of values.
- ✓ **Defn**: Continous data is a numerical data or variable that can assume any numerical value within or over a certain interval (s).
- ✓ Examples of quantitative data are:
  (i) A persons temperature.
  (ii) Yearly, monthly unemployment figures.
  (iii) Number of students in DS 110 in the year 2025.

5.2. QUALITATIVE/CATEGORICAL (NON-NUMERICAL/ATTRIBUTABLE) DATA

- ✓ Qualitative/categorical data are non-numerical data/observations that that can only be classified into groups of categories such as nominal and ordinal scales.
  (i) Nominal data:- This is data with values that are described by words rather than numbers and with no inherent order or rank on a scale.
  (ii) Ordinal data:- This is non-numeric data about order or rank on a scale such as 1,2,3,... or A+, A, B+, B, C+, C, ...
- ✓ In other words, qualitative data are data collected on a qualitative variable.
- ✓ Examples of qualitative or categorical data are:
  (i) Gender type.
  (ii) Blood type.
  (iii) Area of residence.
  (iv) Eye type etc.

NOTE

- ✓ There are other data types beside the quantitative or qualitative data that are classified based on the time over which they are collected. These include, time series data and cross-sectional data.

5.3. TIME SERIES DATA

- ✓ This is a data type where each observation in the sample represents a different equally spaced point in time (e.g., years, months, days).

- ✓ These are data collected on the same element for the same variable at different points in time or for different periods of time .
- ✓ The word **periodicity** in time series simply means the time between observations, it may be annual, quarterly, monthly, weekly, daily, hourly, etc.
- ✓ The data given in the Table below is an example of time-series data. This table lists the average tuition and fee charges in 2014 dollars for the students at four-year public institutions in the United States for selected years.

Years	Tutions and fee (dollars)
1974 – 75	2469
1984 – 85	2810
1994 – 95	4343
2004 – 05	6448
2014 – 15	9139

5.4. CROSS-SECTIONAL DATA

- ✓ This is the data type where each observation represents a different individual unit (e.g., a person, firm, geographic area) at the same point in time.
- ✓ In other words, this is data collected on different elements at the same point in time or for the same period of time.
- ✓ In general, cross-section data contain information on different elements of a population or sample for the same period of time.
- ✓ The data given in the Table below indicating the total wealth of each of the eight richest persons in the world is an an example of cross-section data.

Name	Total Wealth (Billions of dollars)
Billgates	79.2
Calos Slim	77.1
Warren Buffett	72.7
Amancio Ortega	64.5
Larry Ellison	54.3
Charles Koch	42.9
David Koch	42.9
Christy Walton	41.7

6. COLLECTION OF DATA OR DATA COLLECTION

- ✓ Once you have defined your variables, you may need to collect data for those variables.
- ✓ Data collection can be defined as the process of counting or enumerating or measuring together with the recording of results. Data is best collected with a purpose.
- ✓ Before data is collected ask these questions.
  (i) What do you want to know?
  (ii) Why do you want to know it?
- ✓ Data collected in a thoughtful and systematic manner are valuable aid to decision making.
- ✓ Data collected can be classified into two types, these are secondary and primary data.

6.1. SECONDRY DATA

- ✓ Secondary data is data which is already collected, processed and used by someone else for his/her own purpose which may not necessarily be the same as for the current researcher.
- ✓ In other words, it is data collected by someone else and is available in published sources such as journals, newspapers, company newsletters and publications made by government agencies.
- ✓ Although it is much faster and cheaper to obtain secondary data, the data might be out dated, and therefore not suitable for the current study (**advantage/disadvantage**).

6.2. PRIMARY DATA

- ✓ This is data which is originally collected by a researcher from the field for the first time and for a specific purpose.
- ✓ In other words, primary data is data collected directly by the investigator or by the organization employing the investigator.
- ✓ Thus, this data is first hand information, and hence is up to date, more reliable and accurate (advantange).
- ✓ However, primary data is usually expensive to collect and requires more time and labour to gather (disadvantage).
- ✓ Primary data is collected by a variety of ways such as:
  (i) Simple observation,
  (ii) Personal interviews,
  (iii) Questionnaires.

7. METHODS OF DATA COLLECTION

- ✓ Sources of data can be classified as primary source and secondary source giving rise to a primary data and secondary data already discussed.
- ✓ There are several methods of data collection other than the ones aforementioned for primary data.
- ✓ However, there are four most common methods, these are:

7.1. CENSUS.

- ✓ A census is a study that obtains data from every member of a population or a census is a process of collecting all information on all elements of the target population.

NOTE

- ✓ In census, the target population is very large so, in practice, a census is rarely taken because it is expensive and time-consuming.
- ✓ In many cases, it is even impossible to identify each element of the target population.
- ✓ In general, a census is a survey that includes every member of the population.
- ✓ As an example, if we collect information on the 2025 incomes of all families in Zambia, it will be referred to as a census.

7.2. SAMPLE SURVEY.

- ✓ A sample survey is a study that obtains data from a subset of a population, in order to estimate population attributes.
- ✓ In other words, a sample survey is the collection of information from the elements of a sample.
- ✓ In general, a sample survey is a survey that includes only a portion of the population.
- ✓ Usually, to conduct a survey, we select a sample and collect the required information from the elements included in that sample.
- ✓ We then make decisions based on this sample information.
- ✓ As an example, if we collect information on the 2025 incomes of 50 families selected from Zambia, it will be called a sample survey.

7.3. EXPERIMENT.

- ✓ An experiment is a controlled study in which the researcher attempts to understand cause-and-effect relationships.
- ✓ The study is "controlled" in the sense that the researcher controls how subjects are assigned to groups and which treatments each group receives.

7.4. OBSERVATIONAL STUDY.

- ✓ Like experiments, observational studies attempt to understand cause-and-effect relationships.
- ✓ However, unlike experiments, the researcher is not able to control how subjects are assigned to groups and/or which treatments each group receives.

8. ORGANIZATION OF DATA

- ✓ How do we summarise or represent data to make it more comprehensible? We can use
  (a) Tables e.g., frequency distribution tables
  (b) Graphs e.g., histograms, bar graphs (bar charts) pie diagrams, frequency polygons, and ogives.
- ✓ In general, the process of organizing data is done through descriptive statistics techniques.
- ✓ As earlier stated, descriptive statistics consists of methods for organizing, displaying and describing data using tables, graphs, and summary measures.

8.1. FREQUENCY DISTRIBUTION AND CUMULATIVE FREQUENCY DISTRIBUTIONS

- ✓ A frequency distribution is a tabular summary of data showing the number (or frequency) of items (or observations) in each of several non-overlapping categories or classes.
- ✓ In other words, the grouping of x_i with their associated frequencies f is called a frequency distribution.
- ✓ Cumulative frequency distribution is a form of frequency distribution that represents the sum of all the frequencies.
- ✓ In other words, cumulative frequency column lists the total of each frequency.
- ✓ A typical frequency distribution and the cumulative frequency distribution of variable X is presented in table below:

Value of variables	frequency	Cumulative Frequency
X_1	f_1	f_1
X_2	f_2	$f_1 + f_2$
X_3	f_3	$f_1 + f_2 + f_3$
X_i	f_i	$f_1 + f_2 + f_3 + f_i$
X_n	f_n	$f_1 + f_2 + f_3 + f_i + f_n$
Total frequency	N	N

- ✓ There are two kinds of frequency distributions, these include **Ungrouped** and **Grouped** frequency distributions.

A. UN GROUPED FREQUENCY DISTRIBUTION

- ✓ Un-grouped frequency distribution is used if the number of distinct values is small

NOTE

(i) Sometimes a relative frequency distribution and a relative cumuative frequency tables are prepared.

(ii) Relative cumulative frequency distribution is the cumulative frequency divided by the total frequency.

(iii) A relative frequency captures the relationship between a frequency and the total number of observations, and it is defined as:

$$Relative\ frequency\ (RF)\ = \frac{frequency}{Total\ frequency}$$

(iv) The percentage frequency is defined as:

$$Percentage\ frequency\ (\%\ F)\ = \frac{frequency}{total\ frequency} \times 100$$

EXAMPLE

For the following data which represent the number of breakdowns for each truch in a fleet of 20 trucks in one year: 6 11 5 1 6 6 7 5 7 6 1 7 5 3 6 3 5 6 7 6

Construct

(a) a frequency distribution

(b) a relative frequency distribution tables

SOLUTION

(a) Frequency distribution table

Number of breakdowns (x)	**Tally mark**	**Frequency (f)**
1	//	2
3	//	2
5	////	4
6	~~###~~ //	7
7	////	4
11	/	1

(b) Relative frequency distribution table

Number of breakdowns (x)	Frequency (f)	Relative frequency (R.f.)
1	2	0.1
3	2	0.1
5	4	0.2
6	7	0.35
7	4	0.2
11	1	0.05
Total	20	1

EXAMPLE

There are 20 students in a class. The teacher, Ms. Jolly, asked the students to tell their favorite subject. The results are as follows: Mathematics, English, Science, Science, Mathematics, Science, English, Art, Mathematics, Mathematics, Science, Art, Art, Science, Mathematics, Art, Mathematics, English, English, Mathematics

Subject	**No. of students**
Art	4
Maths	7
Science	5
English	4

Construct the relative frequency (RF), percentage frequency (%F) and Cumulative frequency.

SOLUTION

Subject	No. of students	RF	% F	Cumulative frequency
Art	4	0.2	20	4
Maths	7	0.35	35	11
Science	5	0.25	25	16
English	4	0.2	20	20
Total	20			

B. GROUPED FREQUENCY DISTRIBUTION

- ✓ A grouped frequency distribution is used if the number of distinct values is large.
- ✓ In grouped frequency distributions, the first real step is to divide the observed range of variable into a suitable number of *class-intervals* and to record the number of observations in each class.
- ✓ Consider the following table of frequencies (values) in the different classes below

Marks (x)	**Number of students (f)**
15 – 19	9
20 – 24	11
25 – 29	10
30 – 34	44
35 – 39	45
40 – 44	54
45 – 49	37
50 – 54	26
55 – 59	8
60 – 64	5
65 – 69	1
Total	250

- ✓ Such a table showing the distributions of frequencies in the different classes is called a **frequency table** and the manner in which the class frequencies are distributed over class intervals is called the **grouped frequency distribution of the variables**.

REMARK

(1) The classes of the type $15-19, 20-24, 25-29$ etc in which both the upper and the lower limits are included are called **inclusive classes**. For example, $20-24$ includes all the values from 20 to 24, both inclusive 'and the classification is termed as **inclusive type classification**.

(2) If we are dealing with continous variables (or data), it is not possible to arrange the data in the class intervals of the above type. For example, if class intervals $15-19$ and $20-24$ represent the distribution of age in years, then the persons with ages between 19 and 20 years are not taken into consideration.

In such a case we form the class intervals as shown below:

Age in years

Below 5

5 or more but less than 10

10 or more but less than 15

15 or more but less than 20

20 or more but less than 25

and soon.

Here all the persons with any fraction of age are included in one group or the other. For practical purpose we re-write the above classes as:

$0-5$
$5-10$
$10-15$
$15-20$
$20-25$

This form of frequency distribution is known as **continuous frequency distribution**. It should be clearly understood that in the above classes, the upper limits of each class are excluded from the respective classes.

Such classes in which the upper limits are excluded from the respective classes and are included in the immediate next class are known as **exclusive classes** and the classification is termed as **exclusive type classification.**

NOTE

(1) For a grouped data, a relative frequency captures the relationship between a class frequency and the total number of observations, it is given by

$$\text{Relative frequency of a class } (RF) = \frac{\text{frequency of a class}}{Total\ frequency}$$

(2) The percentage frequency is given by

$$Percentage\ freq\ of\ a\ class\ (\%\ F)\ =\ \frac{frequency\ of\ a\ class}{Total\ frequency}\times 100$$

(3) Cumulative percentage frequency of a class is given by

$$cumulative\ Percentage\ =\ \frac{cumulative\ frequency\ of\ a\ class}{Total\ frequency}\times 100$$

✓ We consider the following guidelines to construct frequecy distributions tables for grouped data especially for classification:

(i) Determine the number of non-overlapping and mutually exclusive classes.

(ii) Determine the number of exhaustive classes, *i.e.*, each of the given values should be included in one of the classes.

(iii) Determine the number of classes that should neither be too large nor too small (As a general guideline, we recommend using between 5 and 20 classes).

NOTE

The number of classes may be more" than 20 depending upon the total frequency and the details required. But it is desireable that it is not less than 5 since in that case the classification may not reveal the essential characteristics of the population. The following formula may be used to determine an approximate number *k* of classes:

$k = 1 + 3.322 \log_{10} N$, where N is the total frequency.

(iv) Determine the width of each class by the formula:

$$Approximate\ class\ width\ =\ \frac{largest\ data\ value - smallest\ data\ value}{Desired\ Number\ of\ classes\ (k)}$$

NOTE

The computed value for class width should be increase to the next highest whole number so as to ensure that all the classes taken together cover the data. For example, if

$$(a)\ \ class\ width = \frac{47-1}{6} \approx 7.7 \text{ (increase to 8)}$$

$$(b)\ class\ width = \frac{50-30}{5} = 4 \text{ (increase to 5)}$$

Alternatively, the width of an interval (class width) is the difference between the boundaries, i.e., subtract the lower limit of the current class from the lower limit of the next class.

Symbolically,

$$width\ =\ Lower\ limit\ of\ the\ next\ class\ \ Lower\ limit\ of\ the\ current\ class.$$

(v) Determine the class limits (or data range for each class). Here we find

(a) The lower class limit which is the lowest data value that can fit in a class.

(b) The upper class limit which is the highest data value that can fit in a class.

(vi) Determine the class boundaries. In order to obtain the continuous frequency distrinution of class limits, we determine the class boundaries as follows:

(a) For upper class boundaries, add 0.5 units to the upper class limits.

(b) for lower class boundaries, subtract 0.5 units from the lower class limits.

NOTE

1. Class boundaries are half way points of a space between the upper limit of one class and the lower limit of the next class (intervals).
2. Class boundaries are formed to close up the space between the upper limit of one class and the lower limit of the next class.

EXAMPLE

Construct a grouped frequency distribution table for the following data, which represent the scores of 40 students in a statistics test:

46 49 50 53 54 55 55 57 58 59 59 60 62

63 64 65 65 66 66 66 66 68 68 69 70 71 71

72 73 73 73 74 76 78 79 80 82 83 84 88

Grouped frequency distribution table.

SOLUTION

Score	Tally	Frequency (f)
45 – 49	//	2
50 – 54	///	3
55 – 59	~~////~~ /	6
60 – 64	////	4
65 – 69	~~////~~ ////	9
70 – 74	~~////~~ ///	8
75 – 79	///	3
80 – 84	////	4
85 – 89	/	1

EXAMPLE

The following data were obtained in a survey of the heights of 20 children in a sports club each height was measured to the nearest centimetre. Construct the frequency distribution table for the above data.

133 136 120 138 133 131 127 141 127 143

130 131 125 144 128 134 135 137 133 129

SOLUTION

To form the frequency distributions of the heights of 20 children, group the information into classes or intervals. here are three different ways of writing the same set of intervals

Heights (cm)
$119.5 \leq x < 124.5$
$124.5 \leq x < 129.5$
$129.5 \leq x < 134.5$
$134.5 \leq x < 139.5$
$139.5 \leq x < 144.5$

Height (cm)
$120 - 124$
$125 - 129$
$130 - 134$
$135 - 139$
$140 - 144$

Heights (cm)
$119.5 - 124.5$
$124.5 - 129.5$
$129.5 - 134.5$
$134.5 - 139.5$
$139.5 - 144.5$

NOTE

1. The values 119.5, 124.5,129.5,…are called the class boundaries or the interval boundaries.
2. The upper class boundary (u.c.b) of one interval is the lower class boundary (l.c.b) of the next interval.

The frequency distribution for the above data is given as follows:

Height (cm)	**Tally**	**Frequency (f)**
119.5 – 124.5	/	1
124.5 – 129.5	𝍸	5
129.5 – 134.5	𝍸//	7
134.5 – 139.5	////	4
139.5 – 144.5	///	3
Total		20

4. It is also important to note that when the data is presented only in a form of a grouped frequency distribution, the original information has been lost.

5. For example, you would know that there was an item in the first interval, but you would not know what it was. you would know only that it was between 119.5cm and 124.5cm.

EXAMPLE

Consider the following collected data showing the marks obtained by 40 students in DS 110 test.

64 87 70 55 99 64 89 65 38 78 62 70 67 66 69 39 56 95 75 51 71 99 68 86 53 80 57 50 47 55 81 98 36 83 51 66 63 85 79 70

Group the above data into classes starting with $30-39$, $40-49$, up to $90-99$ in a class width of 10.

SOLUTION

Marks	**Number of students**
$30-39$	3
$40-49$	1
$50-59$	8
$60-69$	10
$70-79$	7
$80-89$	7
$90-99$	4
Total	40

Note that for the class $30-39$

1. 30 is the lower class limit and 39 is the upper class limit
2. $\frac{30+39}{2} = 34.5$ is the midle point class mark or mid-interval value
3. 10 is the class width
4. 29.5 is the lower class boundary, and it is calculated by subtracting half from the lower class limit, 30.
5. 39.5 is the upper class boundary and it is calculated by adding half to the upper class limit 39.

EXAMPLE

Based on the grouped data below. Construct the cumulative frequency distribution.

Time to travel to work	Frequency
1 – 10	8
11 – 20	14
21 – 30	12
31 – 40	9
41 – 50	7

SOLUTION

Time to travel to work	Frequency	Cumulative Frequency
1 – 10	8	8
11 – 20	14	22
21 – 30	12	34
31 – 40	9	43
41 – 50	7	50
Total	50	

8.2. GRAPHICAL REPRESENTATION OF DATA

- ✓ Graphic representations are in a sense pictures of data and as such sometimes make the data easier to comprehend.
- ✓ In other words, a graphical representation is a visual display of data and statistical results.
- ✓ It exhibits the relation between data, ideas, information and concepts in a diagram.
- ✓ It is easy to understand and it is one of the most important learning strategies.
- ✓ It always depends on the type of information in a particular field or area.
- ✓ A graph is a chart through which data are represented in the form of lines or curves drawn across the coordinated points plotted on the surface.
- ✓ Graphs enable us in studying the relationship between two variables.
- ✓ It also helps us to measure the extent of change in one variable when another variable change by a certain amount.
- ✓ There are a number of different graphical representations commonly used for summarizing and organizing data , some of these include:

(1) Line Graphs – Line graph or the linear graph is used to display the continuous data and it is useful for predicting future events over time.

(2) Bar Graphs – Bar Graph is used to display the category of data and it compares the data using solid bars to represent the quantities.

(3) Histograms – The graph that uses bars to represent the frequency of numerical data that are organised into intervals. Since all the intervals are equal and continuous, all the bars have the same width, etc

(4) Circle Graph – Also known as the pie chart that shows the relationships of the parts of the whole. The circle is considered with 100% and the categories occupied is represented with that specific percentage like 15%, 56%, etc.

(5) Stem and Leaf Plot – In the stem and leaf plot, the data are organised from least value to the greatest value. The digits of the least place values from the leaves and the next place value digit forms the stems.

(6) Box and Whisker Plot – The plot diagram summarizes the data by dividing into four parts. Box and whisker show the range (spread) and the middle (median) of the data.

8.2.1. CUMULATIVE FREQUENCY CURVE (OGIVE)/ RELATIVE CUMULATIVE FREQUENCY CURVE

- ✓ A cumulative frequency curve or ogive is a curve obtained by plotting the graph of the cumulative frequencies against the number of data values.
- ✓ A cumulative frequency curve takes the form of an elongated **S.**
- ✓ A relative cumulative frequency curve is similar to a cumulative curve, but instead of plotting cumulative frequencies, we plot relative cumulative frequencies against the number of data values.

NOTE

The main purpose of the two curves is to estimate quartiles and percentiles.

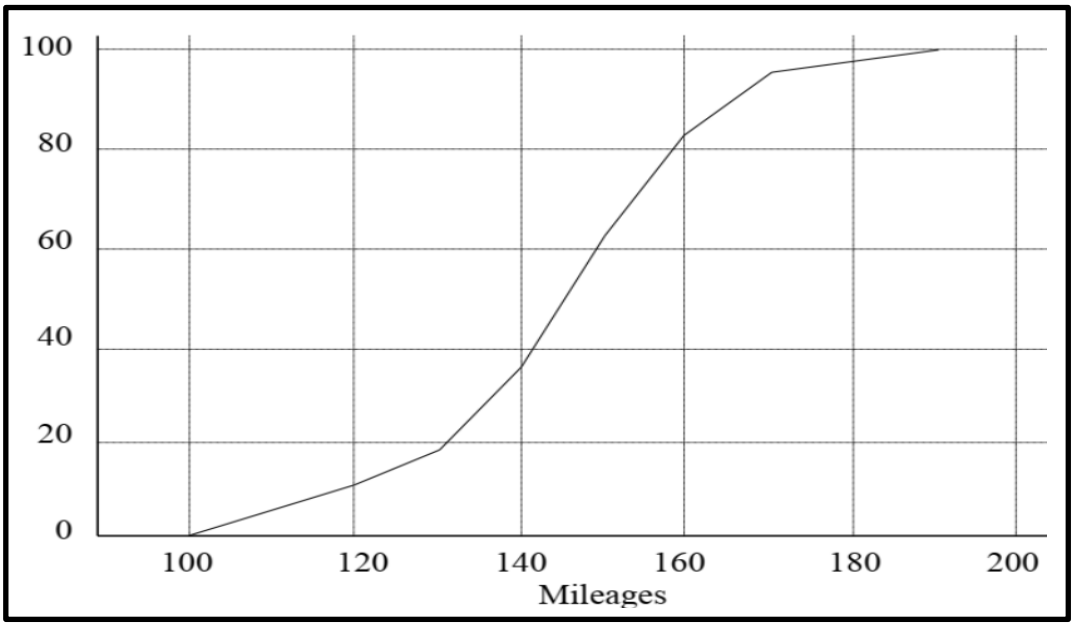
EXAMPLE

Use the cumulative frequency table below to construct the cumulative frequency curve.

Interval less than	Frequency	Cumulative frequency	% cumulative frequency
100	0	0	0.0
120	4	4	10.0
130	3	7	17.5
140	7	14	35.0

150	11	25	62.5
160	8	33	82.5
170	5	38	95.0
190	2	40	100.0

SOLUTION

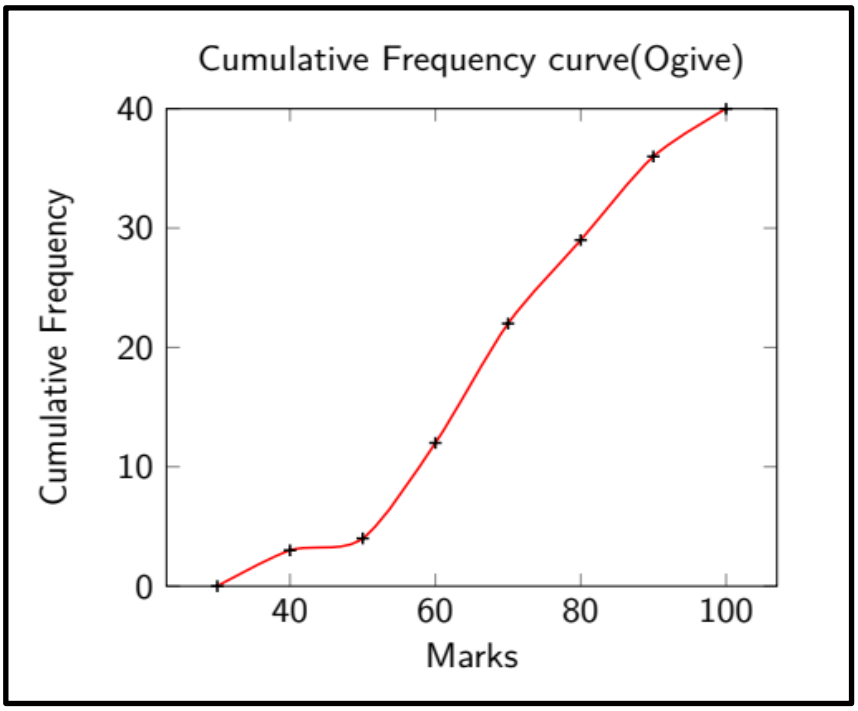


EXAMPLE

Using the cumulative frequency distribution table given below, construct the cumulative frequency curve or ogive.

Marks	**30**	**40**	**50**	**60**	**70**	**80**	**90**	**100**
Cumulative Frequecy	0	3	4	12	22	29	36	40

SOLUTION



8.2.2. HISTOGRAM

- ✓ A histogram is also a graphical representation of a frequency distribution but using two axes, i.e., the X and Y axes.
- ✓ Thus, a working definition of a histogram is that it is a bar chart for grouped numerical data in which we use vertical bars to represent the frequencies or percentages in each group.
- ✓ In a histogram, there are no gaps between adjacent bars (except when there are no data in a particular bar).
- ✓ You display the variable of interest along the horizontal (X) axis.
- ✓ The vertical (Y) axis represents either the frequency or the percentage of values per class interval.
- ✓ Thus, in a histogram rectangles are drawn so that the area of each rectangle is proportional to the frequency in the range covered by it.

NOTE

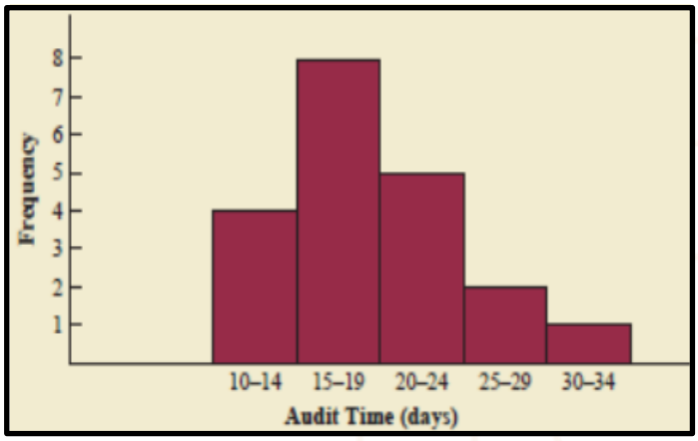
Grouped data can be displayed in a histogram.

EXAMPLE

Using the data displayed in the frequency distribution table below, construct the histogram

Audit time (days)	Frequency
10 – 14	4
15 – 19	8
20 – 24	5
25 – 29	2
30 – 34	1
Total	**20**

SOLUTION

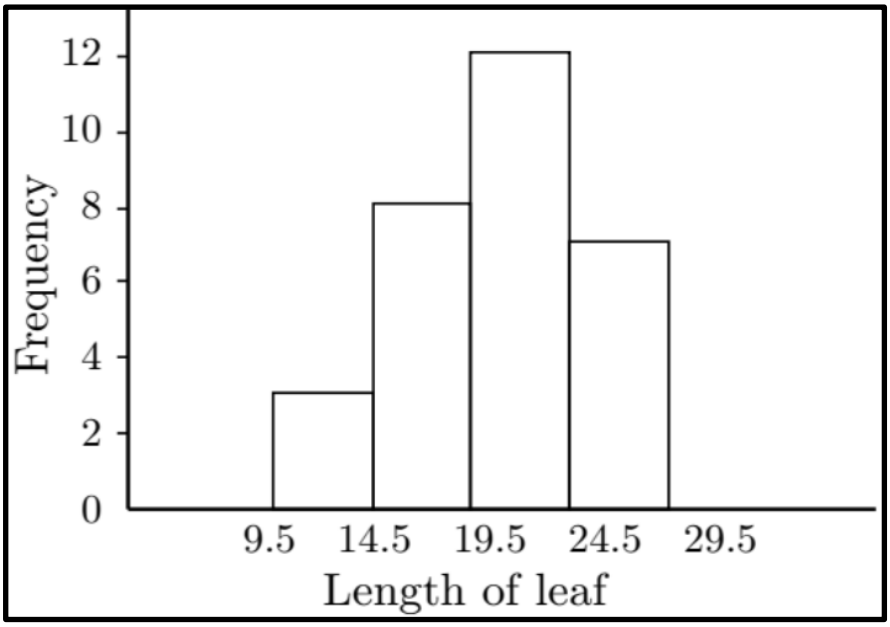


EXAMPLE

The lengths of 30 plant leaves of species were measured and the information were grouped as shown in the table below. Measurements were taken correct to the nearest cm. Draw a histogram to illustrate the data given in a frequency distribution table below.

Length of leaf (cm)	9.5−14.5	14.5−19.5	19.5−24.5	24.5−29.5
Frequency	3	8	12	7

SOLUTION

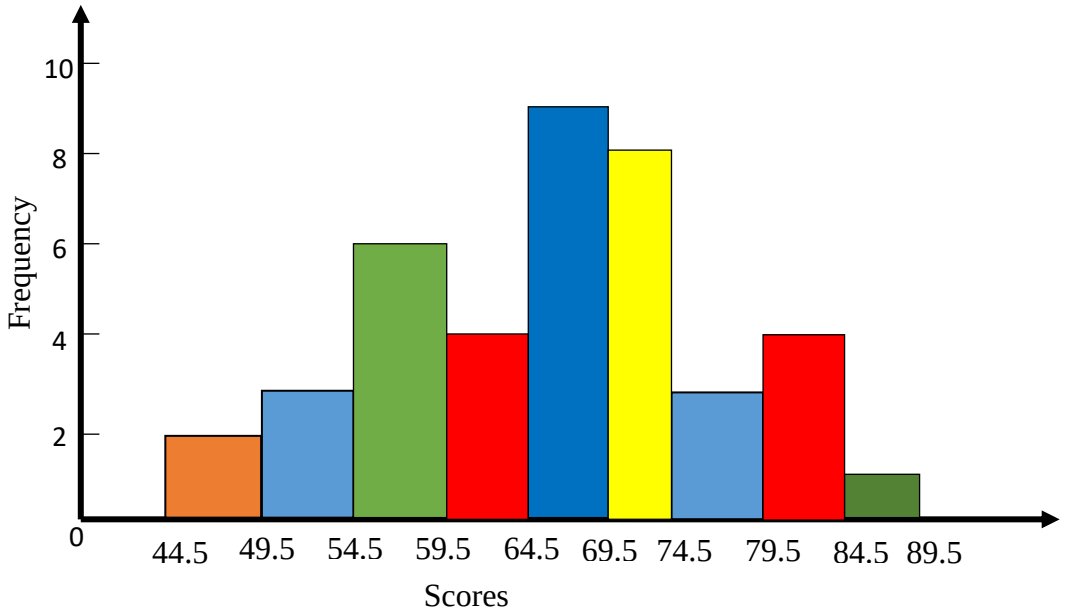


EXAMPLE

Construct a frequency histogram for the grouped frequency distribution of the 40 statistics test scores.

Marks	**Number of students**
30 – 39	3
40 – 49	1
50 – 59	8
60 – 69	10
70 – 79	7
80 – 89	7
90 – 99	4
Total	40

SOLUTION



NOTE

1. The class boundaries on the histogram are halfway between the class intervals.
2. Sometimes histograms are presented on the relative frequency scale resulting in a relative frequency histogram.

8.2.3. BAR CHART (OR GRAPH)

- ✓ A bar chart compares different categories by using individual bars to represent the tallies for each category.
- ✓ A bar graph is used to display categorical data, implying that a bar graph is most appropriate for qualitative (non numerical) variables.
- ✓ For each value or grouping of a variable, a rectangle is drawn with a base of constant width and height proportional to the frequency.
- ✓ The rectangles are then displayed (equally spaced from one another) not to provide distortions.
- ✓ The length of a bar represents the amount, frequency, or percentage of values falling into a category.

GENERAL GUIDELINES WHEN CONSTRUCTING A BAR CHART

1. Label the Frequencies vertically
2. Locate the categories of the variable on a horizontal scale.
3. Label the frequencies along the vertical axes.
4. Use whatever scale is considered appropriate.
5. Construct a rectangle over each category of the qualitative variable with the height equal to the number of the variable in the category.

NOTE

A bar graph is also used to depict a frequency distribution especially when the data values are non-numerical.

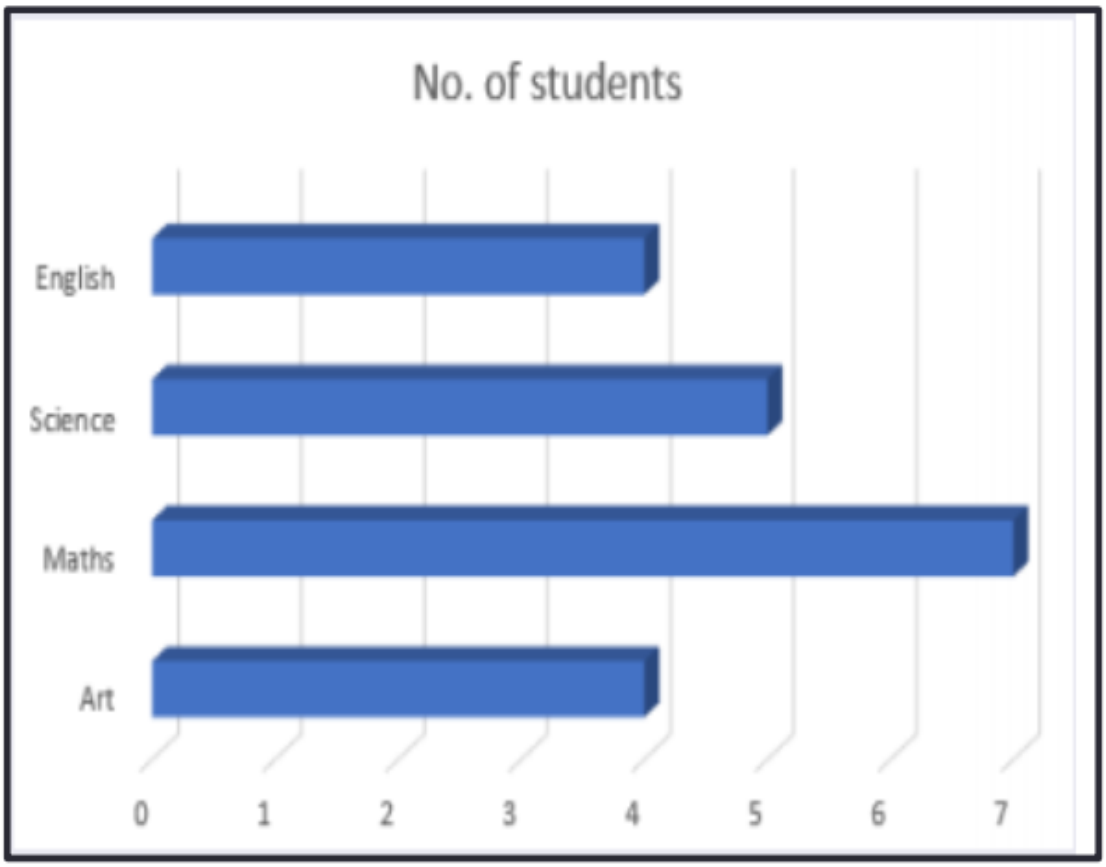
EXAMPLE

There are 20 students in a class. The teacher, Ms. Jolly, asked the students to tell their favorite subject. The results are as follows: Mathematics, English, Science, Science, Mathematics, Science, English, Art, Mathematics, Mathematics, Science, Art, Art, Science, Mathematics, Art, Mathematics, English, English, Mathematics

Subject	No. of students
Art	**4**
Maths	**7**
Science	**5**
English	**4**

Construct a bar chart

SOLUTION

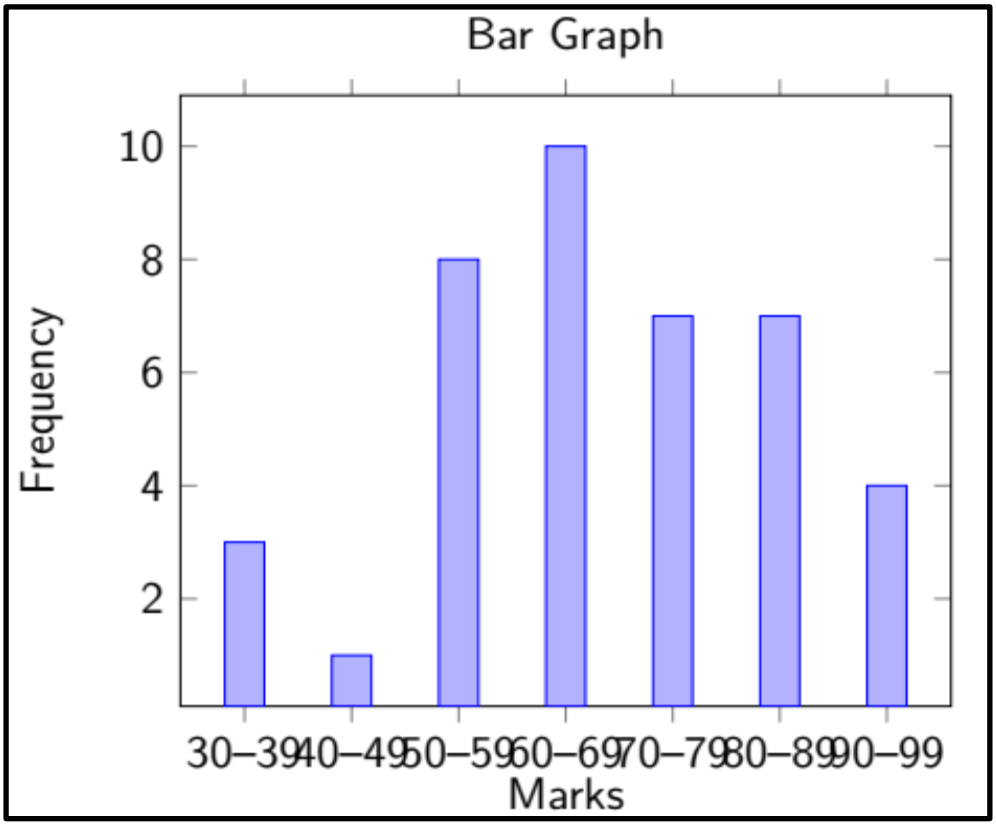


EXAMPLE

Consider the data given in the frequency distribution table below and construct the bar graph representing the data

Marks	Number of students
30 – 39	**3**
40 – 49	**1**
50 – 59	**8**
60 – 69	**10**
70 – 79	**7**
80 – 89	**7**
90 – 99	**4**
Total	**40**

SOLUTION



8.2.4. FREQUENCY POLYGON

✓ Another way of graphically displaying a grouped frequency distribution is by means of a frequency polygon.

- ✓ A frequency polygon is constructed by joining dots that represent the frequency of each group.
- ✓ Sometimes a frequency polygon can be superimposed on the histogram (drawing a histogram and a frequency polygon on the same axes). This is usually happens in grouped data.
- ✓ In general, a grouped frequency distribution can be displayed as a frequency polygon.
- ✓ To construct a frequency polygon for each interval in a grouped data, plot frequency density against the mid interval value, where

$$\text{Mid interval value } = \frac{1}{2}(lower\ class\ boundary + upper\ class\ boundary)$$

$$\Leftrightarrow \text{ Mid interval value } = \frac{1}{2}(lower\ class\ limit + upper\ class\ limit)$$

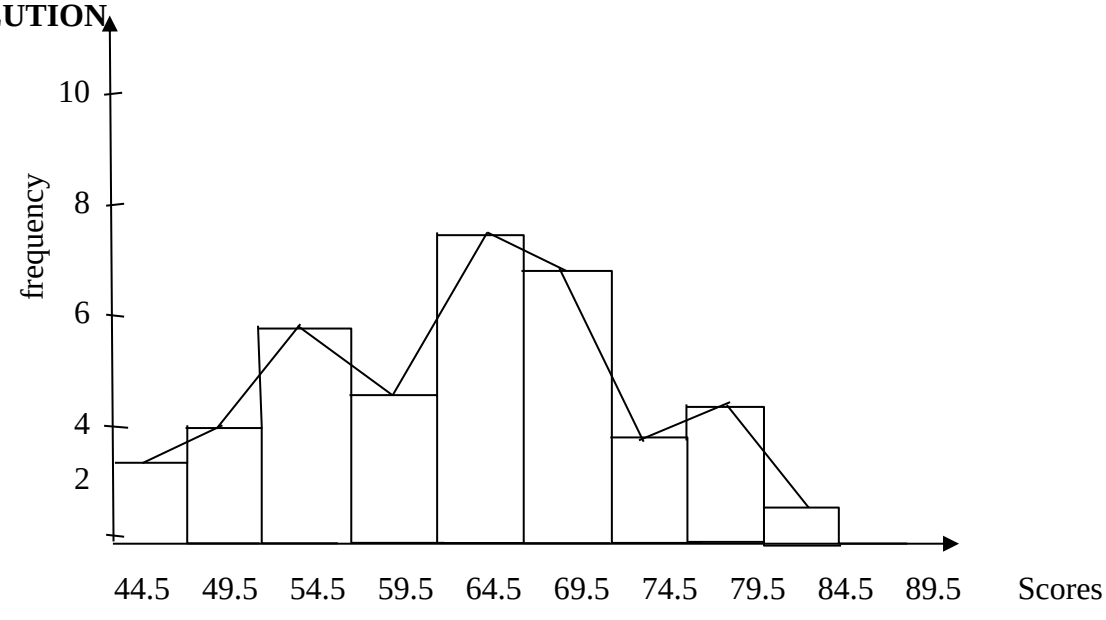
- ✓ Then later join the points with straight lines.

EXAMPLE

Construct a frequency polygon on the same axes as a histogram for the statistics test scores

Marks	**Number of students**
$30-39$	3
$40-49$	1
$50-59$	8
$60-69$	10
$70-79$	7
$80-89$	7
$90-99$	4
Total	40

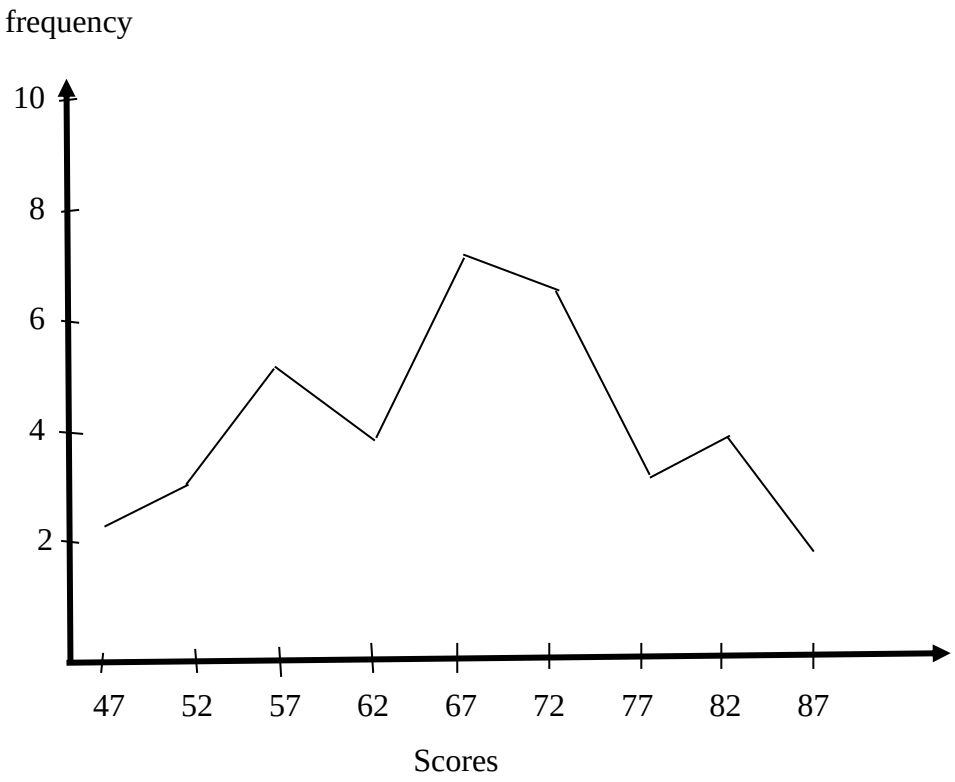
SOLUTION



- ✓ like earlier stated , a frequency polygon can also be constructed on its own by joining dots on the midpoints of each class, the midpoints for the above example are

Class	Midpoint
45 – 49	47 i.e (45+49)/2
50 – 54	52
55 – 59	57
⋮	⋮

frequency polygon

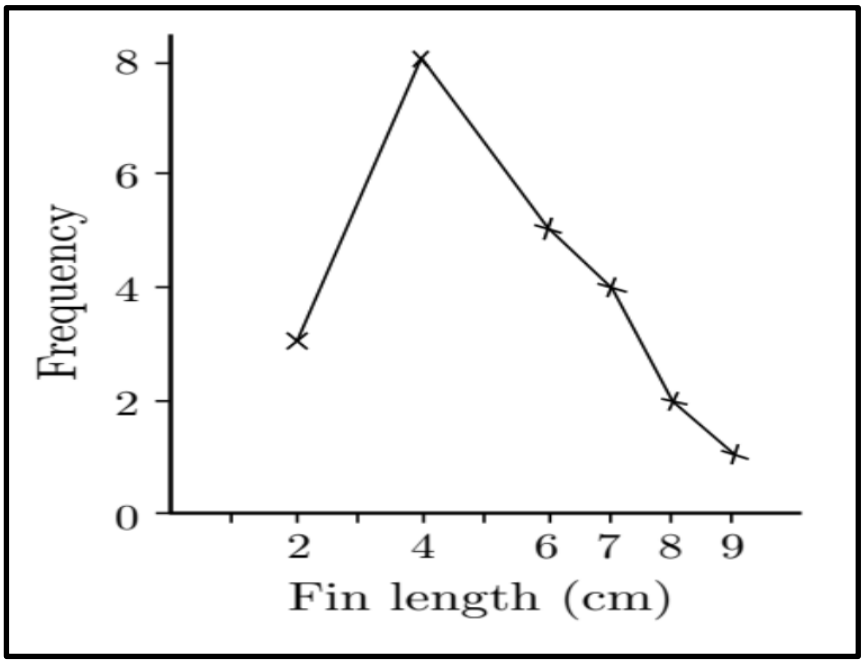


EXAMPLE

Ungrouped data*:* The fin length in (cm) of particular type of fish is given. Draw a frequency polygon to illustrate this information

Fin length (cm)	**2**	**4**	**6**	**7**	**8**	**9**
Frequency	3	8	5	4	2	1

SOLUTION

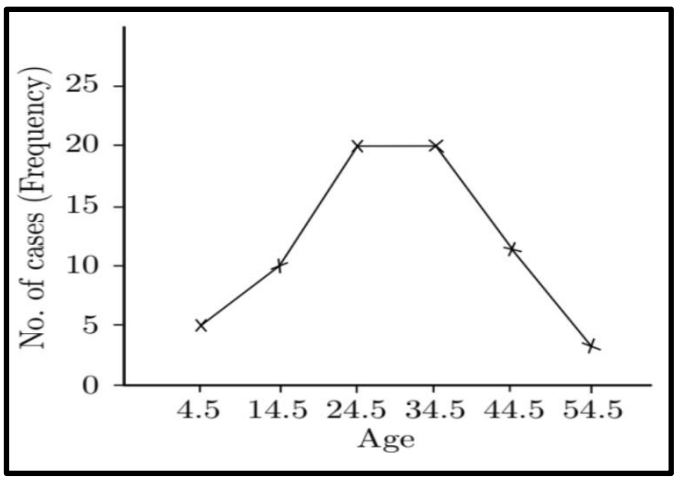


EXAMPLE

Grouped data: The following table shows the age distribution of cases of a certain disease reported during a year in a particular state. Prepare a frequency polygon

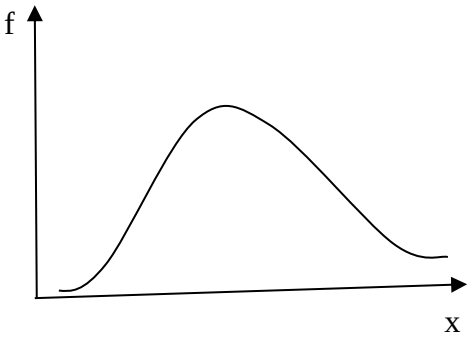
Age	Number of cases
5 – 14	5
15 – 24	10
25 – 34	22
35 – 44	22
45 – 54	13
55 – 64	5

SOLUTION

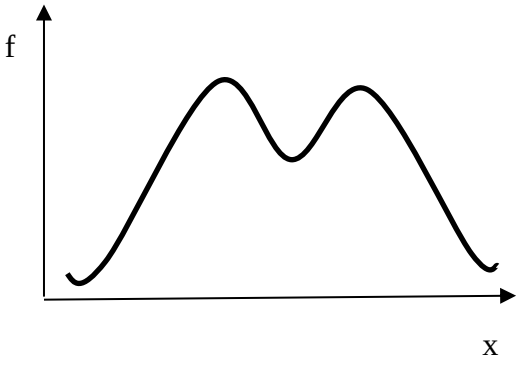


NOTE

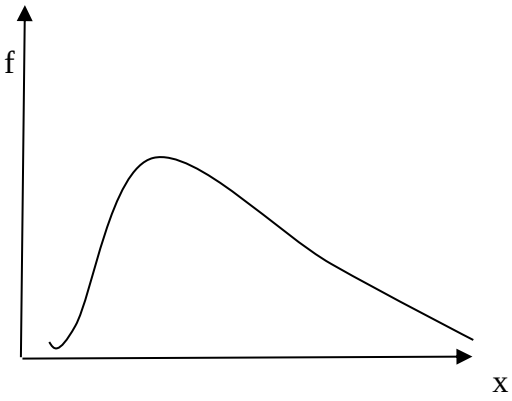
1. A frequency polygon can also be constructed on a relative frequency scale giving a relative frequency polygon.
2. A frequency polygon is useful in conveying the shape of the distribution.
3. The shapes of frequency polygons (or frequency histograms) often resemble commonly observed types:



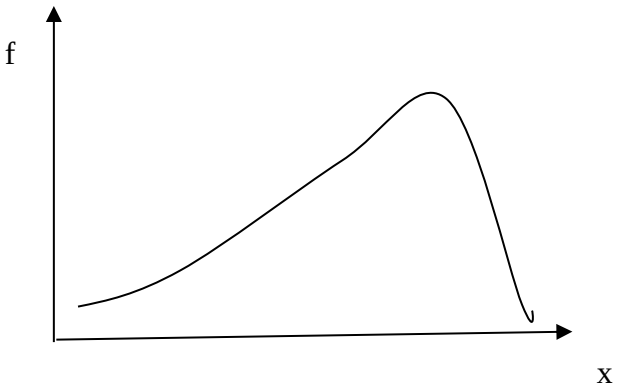
(a) Bell-shaped



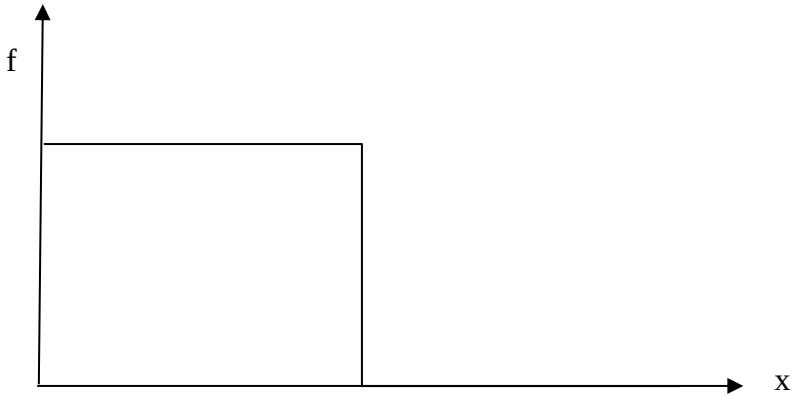
(b) Bimodal (having two modes)



(c) Skewed to the right



(d) Skewed to the left



(e) Uniform

Distributions in (a) and (b) are symmetric. These distributions are often encountered when the data or x values represent test scores. The distribution in (c) is common for variables like income. The distribution in (d) is common for variables like height of people (children and adults). The distribution in (e) might represent the number of phone calls arriving at a switchboard.

8.2.5. STEM AND LEAF PLOTS

- ✓ Stem and leaf plots are a way of organizing data in such a way that the data values themselves are used to construct the bars of a figure that resembles a histogram.
- ✓ It is a very useful way of grouping data into classes while still retaining the original data.
- ✓ This display organizes data into groups, that is, the stems and the leaves.
- ✓ It is done by separating the values into stem digits and leaf digits.
- ✓ For instance, the observation 75 has a stem value of 7 and a leaf of 5.
- ✓ The plot portrays the shape of the distribution and gives the individual data values as well.
- ✓ In general, a stem-and-leaf display allows you to see how the data are distributed and where concentrations of data exist.

EXAMPLE

Construct a stem and leaf plot for the following data:

27 32 47 54 36 44 56 49 53 51
35 29 62 19 65 37 48 33 41 45

SOLUTION

Stem and Leaf plot

1	9
2	7 9
3	2 6 5 7 3
4	7 4 9 8 1 5
5	4 6 3 1
6	2 5

or

Ordered Stem and Leaf plot

1	9
2	7 9
3	2 3 5 6 7
4	1 4 5 7 8 9
5	1 3 4 6
6	2 5

Key: 2|7 = 27

The distribution of the data looks bell-shaped.

EXAMPLE

The maximum temperature in ℃, measured to the nearest degree,was recorded each day and the results are as shown in the table below:

19 23 19 19 20 12 19 22 22 16 18 16 19 20 17

13 14 12 15 17 16 17 19 22 22 20 19 19 20 20

Draw a stem leaf diagram to illustrate the temperatures and write down the modal temperature.

SOLUTION

The smallest value is 12 and the largest value is 23. Thus, grouping the data into intervals$10-19, 20-29, \ldots$ will give you very litle information. Hence, we choose sensible number of intervals usually between 5 and 10. Since you must use intervals with equal width, you could use intervals of 2℃ and consider $12-13,\ 14-15,$

$16-17,\ 18-19,\ 20-21,\ 22-23.$

Stem	Leaf
1	2 2 3
1	4 5
1	6 6 6 7 7 7
1	8 9 9 9 9 9 9 9
2	0 0 0 0 0
2	2 2 2 2 3

Key: 1|2 = 12℃

NOTE

The stem does not necessarily represents the tens digits. For example suppose you want to use the intervals $12-14,\ 15-17,\ 18-20,\ 21-23$. The intervals, $18-20$ can not be represented by a stem of 1, since the tens digits changes during the interval. Thus, for the stem you can use 12, 15, 18 and 21. The leaf is then given as the number that is added to the stem.

Stem	Leaf
12	0 0 1 2
15	0 1 1 1 2 2 2
18	
21	0 1 1 1 1 1 1 1 1 2 2 2 2 2 1
	1 1 1 2

Key: 15|2 = 17℃

18|0 = 18℃

EXAMPLE

The table gives the number of days on which rain fell in 36 consecutive intervals of 30 days

21	19	6	12	8	18	9	8	11	17	15	13
16	9	17	18	9	24	17	7	8	17	17	8
7	11	16	17	8	5	13	22	20	16	20	
13											

Draw stem and leaf diagrams with the following class intervals

(a) $5-9, 10-14, 15-19, 20-24$

(b) $4-6,\ 7-9,\ 10-12,\ 13-15,\ 16-18,\ 19-21,\ 22-24$

SOLUTION

(a)

Stem	Leaf
0	5 6 7 7 8 8 8 8 8 9 9 9
1	1 1 2 3 3 3
1	5 6 6 6 7 7 7 7 7 7 8 8 9 9
2	0 0 1 2 4

Key: 1|6 = 16

NOTE

The Stem and Leaf diagram could have been written differently as follows:

Stem	Leaf
5	0 1 2 2 3 3 3 3 3 4 4 4
10	1 1 2 3 3 3
15	0 1 1 1 2 2 2 2 2 3 3 4
20	0 0 1 2 4

Key: 15|1 = 16; 5|3 = 8

(b) using interval $4-6, 7-9, 10-12$, ... the completed diagram, arranged in order is

Stem	Leaf
4	1 2
7	0 0 1 1 1 1 1 2 2 2
10	1 1 2
13	0 0 0 2
16	0 0 0 1 1 1 1 1 1 2 2
19	0 1 1 2
22	0 1

Key: 13|2 = 15

NOTE

Both diagrams show that the mode is 17 rainy days but the seven intervals used in (b) show more clearly the two peaks, illustrating that the distribution is approximately bi-modal, with modal classes $7-9$ and $16-18$.

BACK-TO- BACK STEMPLOTS

Stem and leaf diagrams can be used to compare two samples by showing the results together on a back-to-back stemplot.

EXAMPLE

Use the stem and leaf diagram to compare the examination marks in French and English for a class of 20 pupils.

French	75	69	58	58	46	44	32	50	53	78
	81	61	61	45	31	44	53	66	47	57
English	52	58	68	77	38	85	43	44	56	65
	65	79	44	71	84	72	63	69	72	79

SOLUTION

French		English
1 2	3	8
7 6 5 4 4	4	3 4 4
8 8 7 3 3 0	5	2 6 8
9 6 1 1	6	3 5 5 8 9
8 5	7	1 2 2 7 9 9
1	8	4 5

Key: 8|5 = 58 Key: 6|3 = 63

Conclusion: From the diagram it is clear that the class had higher marks in English than in French and it appears that they performed better in English. This would however depends on the standards of marking used in the two examinations

NOTE

(1) Each leaf consists of a single digit.

(2) A stem can have any number of digits e.g. $112 \longrightarrow 11|2$, $32 \longrightarrow 3|2$ etc.

PURPOSE OF A STEM AND LEAF DISPLAY

The purposes of a stem and leaf display include the following:

(1) To show the shape of the distribution.

(2) To sort the observations that make up the distribution in order of size.

(3) To make it easy to detect unusual observations and observations which appear more frequent than the others.

(4) To provide a basis for judging the suitability of different types of averages

ADVANTAGE OF STEM AND LEAF DIAGRAM

(1) No information on the value of each observation or original data value is lost

(2) It provides the essential features of a histogram.

DISADVANTAGE OF STEM AND LEAF DIAGRAM

(1) The stem and leaf plot becomes difficult if the number of observations is large or if the data are expressed to more than two significant figures. e.g. 594.

(2) Sometimes it is difficult to tell which value is an outlier.

8.2.6. PIE DIAGRAMS (OR PIE CHARTS)

- ✓ **Defn**: A pie chart is a circle divided into portions that represent the relative frequencies or percentages of a population or a sample belonging to different categories.
- ✓ In other words, a pie chart uses parts of a circle to represent the tallies of each category.

- ✓ The size of each part, or pie slice, varies according to the percentage in each category.
- ✓ Thus, the pie chart in most cases displays total percentages or numbers of observations falling into each of the categories of the qualitative variables presented in form of a circle.
- ✓ The following are the guidelines needed when constructing a pie chart.

(1) Ideally choose a small number of categories, preferably use up to a maximum of six categories.

(2) Then compute degrees by dividing the number of observations (measurements) in categories by the total number of observations (measurement) then multiply by 360º.

(3) Always give a title to the Pie Chart.

COMMON ERRORS IN PIE CHART USAGE

(1) Pie charts can only convey a general idea of the data values.

(2) Pie charts are ineffective when they have too many slices.

(3) Pie chart data must represent parts of a whole (e.g., percent market share).

EXAMPLE

There are 20 students in a class. The teacher, Ms. Jolly, asked the students to tell their favourite subject. The results are as follows: Mathematics, English, Science, Science, Mathematics, Science, English, Art, Mathematics, Mathematics, Science, Art, Art, Science, Mathematics, Art, Mathematics, English, English, Mathematics.

Subjects	Number of Students
Art	4
Maths	7
Science	5
English	4

Construct a pie chart.

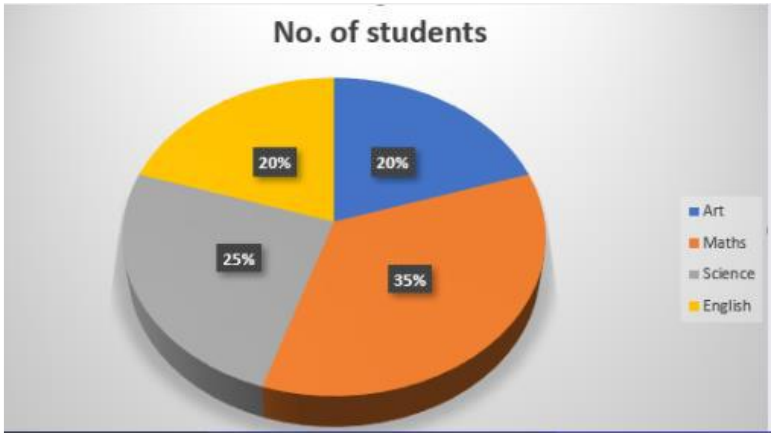
SOLUTION

Art: $\frac{4}{20} \times 100 = 20\% \Leftrightarrow \frac{4}{20} \times 360° = 72°$

Maths: $\frac{7}{20} \times 100 = 35\% \Leftrightarrow \frac{7}{20} \times 360° = 126°$

Science: $\frac{5}{20} \times 100 = 25\% \Leftrightarrow \frac{5}{20} \times 360° = 90°$

English: $\frac{4}{20} \times 100 = 20\% \Leftrightarrow \frac{4}{20} \times 360° = 72°$



EXAMPLE

A sample of 30 persons who often consume donuts were asked what variety of donuts is their favourite. The responses from these 30 persons are as follows:

Glazed, filled, other, plain, glazed, other, frosted, filled, filled, glazed, other, frosted, glazed, plain, other, glazed, glazed, filled, frosted, plain, other, other, frosted, filled Filled, other, frosted, glazed, glazed, filled.

(a) Determine the frequency, relative frequency and percentage distributions for these Data.

(b) Construct the pie chart for the percentage distribution.

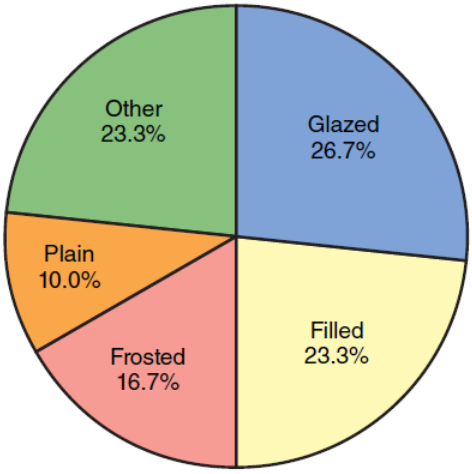
(c) Construct the bar chart.

SOLUTION

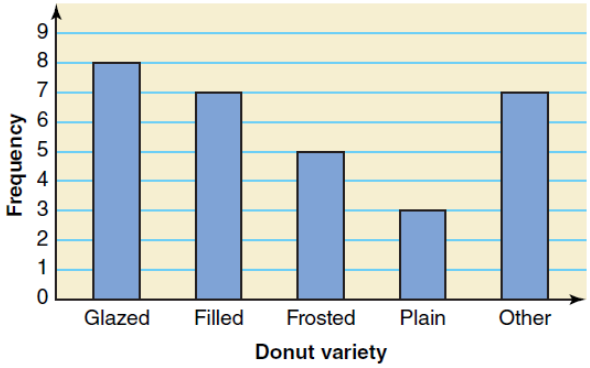
(a) The table below shows the frequency, relative distributions and percentage distributions for the data.

Donut Variety	**Tally**	**Freq (f)**	**Relative frequency**	**percentage**
Grazed	𝍸 ///	8	$8/30 = .267$	$.267(100) = 26.7$
Filled	𝍸 //	7	$7/30 = .233$	$.233(100) = 23.3$
Frosted	𝍸	5	$5/30 = .167$	$.167(100) = 16.7$
Plain	///	3	$3/30 = .100$	$.100(100) = 10.0$
Other	𝍸 //	7	$7/30 = .233$	$.233(100) = 23.3$
Total		Sum = 30	Sum = 1.00	Sum = 100%

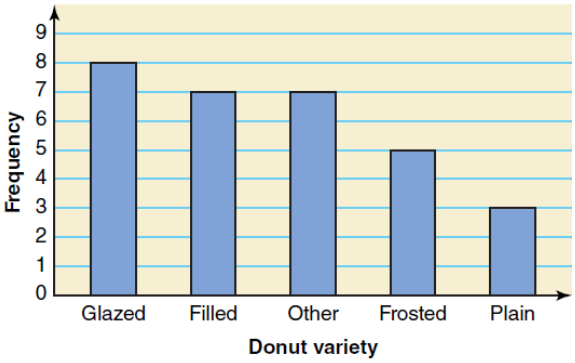
(b) The diagram below shows the pie chart for the percentage distribution provided in the table above.



(c) The diagrams below show the bar chart for the frequency distribution of displayed in the table above.



(a)



(b)

NOTE

(1) The bar chart (b) is called a **Pareto bar chart** which is a bar graph with bars arranged by their heights in descending order.

(2) To make a Pareto chart, arrange the bars according to their heights such that the bar with the largest height appears first on the left side, and then subsequent bars are arranged in descending order with the bar with the smallest height appearing last on the right side.